Money and Success in Football

Does Financial Strength Predict League Position?

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Does Financial Strength Predict League Position?

Research Question: Does a squad's financial rank within its league systematically determine where it finishes in the final standings?

Why this matters:

- Europe's five major leagues generate over **€18 billion** in annual revenue; transfer spending exceeded **€12 billion** in 2023–24 alone.
- UEFA introduced **Financial Fair Play** precisely to limit financial dominance — implying money *does* buy success.
- If outcomes are financially predetermined, competitive drama is fundamentally reduced for fans of smaller clubs.

This study uses 975 team-seasons from 5 leagues and 10 seasons (2015–16 to 2024–25) to answer with evidence, not opinion.

Two contrasting cases — one financially

expected, one statistically miraculous — motivate why the question requires careful empirical investigation.

Two Cases: One Expected, One Miraculous

Leicester City 2015–16

Financial Rank	**11th** of 20
League	Premier League
Squad value	£161M (avg: £458M)
Model prediction	11th place
Actual finish	**1st (Champion)**
Residual	$-$**10.5** positions
Bookmaker odds	**5 000-to-1**

The largest single-season overperformance in the entire 10-year dataset — a statistical shock that occurs with probability well below 0.1%.

Bayer Leverkusen 2023–24

Financial Rank	**2nd** (Bundesliga)
Result	**Bundesliga champions**
Historic feat	First unbeaten Bundesliga season
Context	Ended Bayern's 11-year streak
Residual	-1.8 (within normal range)

Remarkable by football standards, but statistically *expected*: the second-richest squad winning is precisely what the financial hierarchy predicts.

These two cases bracket the full spectrum of outcomes: from the financially-expected champion to the greatest financial upset in modern football.

Data and Scope: 975 Team-Seasons Across Five Leagues

Two data sources:

- **ESPN FC** — final league standings, all five leagues, seasons 2015–16 through 2024–25.
- **Transfermarkt** — squad total market value (€M) at the start of each season.

League	N	Mean MV	r(FR,Pos)
Bundesliga	180	€237M	0.783
La Liga	200	€257M	0.787
Ligue 1	196	€185M	0.791
Premier League	200	€458M	0.801
Serie A	199	€247M	0.889
All	**975**	€278M	**0.813**

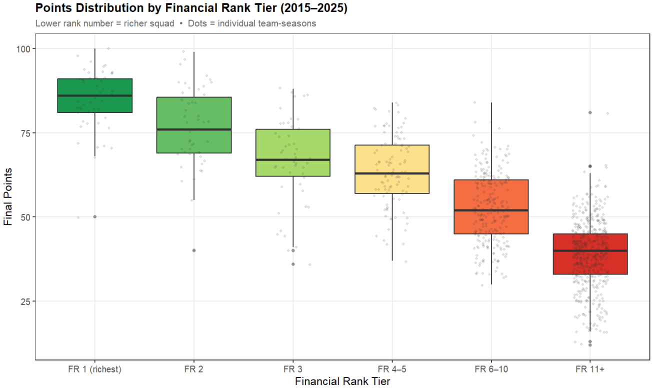
All r values: $p < 0.001$.

Key Variable: Financial Rank (FR)

- Squads ranked by market value **within each league-season**: $\text{FR} = 1$ richest, $\text{FR} = n$ least wealthy.
- **Why rank, not raw euros?** A €300M squad is the *richest* in Ligue 1 but only the *fifth-richest* in the Premier League. Ranking within leagues removes this scale problem.
- $\text{FR} = 1$ always means “the wealthiest club in this league this season,” regardless of whether that squad is worth €600M or €150M.

Even before any model: correlations above 0.8 tell us money is a very powerful predictor. Pearson $r = 0.813$, Spearman $\rho = 0.814$; both $p < 0.001$, $N = 975$.

Financial Rank Creates a Balanced, Meaningful Hierarchy

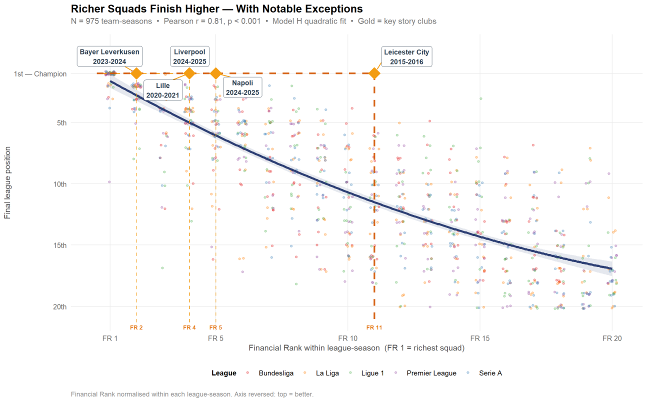


Why Uniformity Matters

- Every league-season contains exactly as many clubs as rank positions, so each FR value appears equally often by construction.
- FR = 1 appears exactly 50 times (once per league-season), as do FR = 2, FR = 3, and so on.
- No structural concentration at any particular rank: the system is **clean and balanced**.

The boxplot confirms a monotonic decline in league points as Financial Rank increases — the

Richer Squads Finish Higher — Across All 975 Team-Seasons



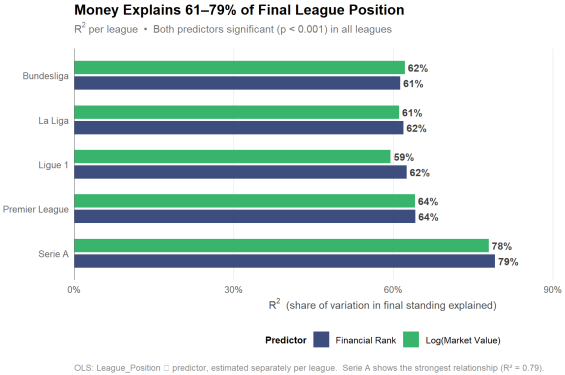
$r = 0.813 \quad R^2 = 0.671$

Pearson correlation and variance explained

FR vs. Final Position, $N = 975$, $p < 0.001$

- FR = 1–2 clubs cluster strongly near the top of their league.
- FR $\geq$ 15 clubs cluster near the bottom.
- The fitted curve (Model H) bends at the bottom: diminishing returns.

The Financial Hierarchy Holds in Every League



League	r	R^2
Bundesliga	0.783	0.613
La Liga	0.787	0.619
Ligue 1	0.791	0.625
Premier League	0.801	0.641
Serie A	**0.889**	**0.791**
All (pooled)	0.813	0.671

Financial Rank alone explains 61–79 % of final position *within each individual league*. Serie A shows the strongest relationship ($R^2 = 0.79$); even the weakest (Bundesliga, $R^2 = 0.61$) is very high by

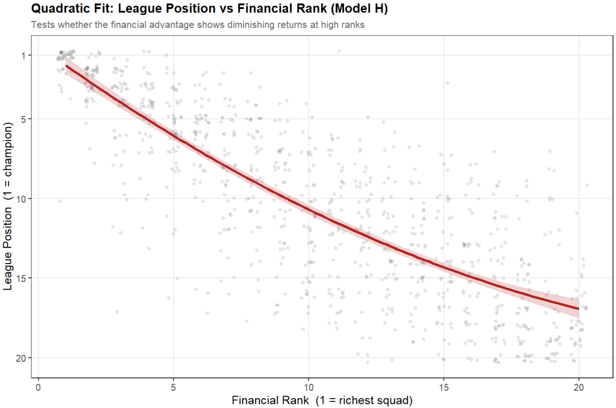
Model Selection: AIC Identifies Model H as Best Specification

Mdl	Formula	R^2	Adj. R^2	AIC
A	FR only (linear baseline)	0.660	0.660	5099.8
C	Log(Market Value) only	0.533	0.532	5410.6
H	**$\mathbf{FR + FR^2}$ (quadratic)**	**0.671**	**0.670**	**5070.5**
H4	$\mathrm{FR} + \mathrm{FR}^2 + \mathrm{League} + \mathrm{Season}$	0.671	0.669	5079.3
J	Log(FR)	0.626	0.625	5194.0
E	Champion $\sim$ FR (Logit)	McFadden $= 0.548$		182.4

Model H wins on AIC: adding one extra term ($\mathbf{FR^2}$) lowers AIC by 29.3 points vs. the linear baseline. Extended Model H4 (league + season controls) scores higher AIC than H — the controls add no explanatory power ($F(5) = 0.247$, $p = 0.941$).

OLS response: League Position ($1 =$ champion), $N = 975$. Lower AIC $=$ better fit per parameter. Bold $=$ preferred specification. Model E uses logistic regression for the binary outcome (champion vs. not).

Model H: Diminishing Returns to Financial Advantage



Model H Equation

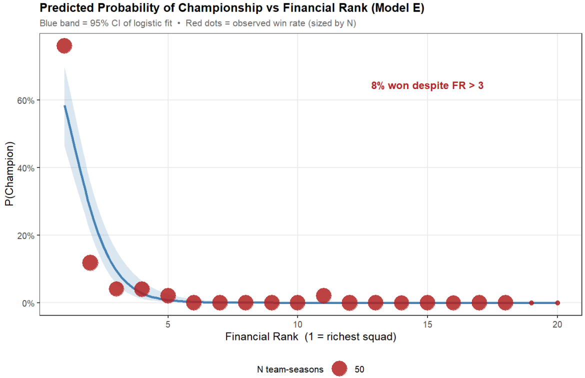
$$\widehat{\text{Pos}} = 0.42 + 1.23\,\mathrm{FR} - 0.020\,\mathrm{FR}^2$$

- $\hat{\beta}_1 = 1.231$ (SE 0.077), $p < 0.001$
- $\hat{\beta}_2 = -0.020$ (SE 0.004), $p < 0.001$
- $R^2 = 0.671$

Marginal cost of one rank step

- FR $1 \to 2$: $\approx$ **1.19** positions lost
- FR $15 \to 16$: $\approx$ **0.63** positions lost

Championship Probability Plummets With Financial Rank



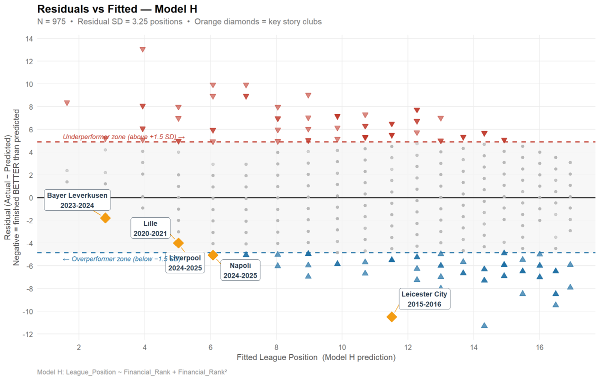
FR	P(Champion)
1 (richest)	**58.5 %**
2	27.8 %
3	9.5 %
5	0.8 %
11	<0.01 %

Model E:

Champion ∼ FR (logistic). $\hat{\beta}_{\mathrm{FR}} = -1.298$, $p < 0.001$. Each rank step multiplies odds by 0.273 (−72.7 %). McFadden $R^2 = 0.548$.

FR = 1: 58.5 % title probability — ≈3× the baseline for a 20-team league. This is structural advantage, not certainty: winning from FR = 11 is virtually impossible by this model.

Who Beat the Model? Residual Analysis



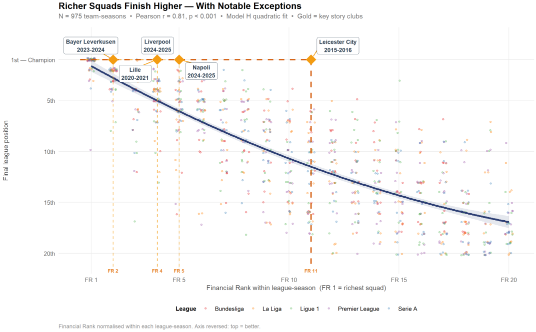
What Is a Residual?

Residual = Actual − Predicted

- **Negative**: finished **better** than predicted (**overperformer**)
- **Positive**: finished **worse** than predicted (**underperformer**)
- Near zero: correctly predicted

Dashed lines mark ±1.5 SD (≈ ±4.9 positions). Teams outside this band are genuine statistical outliers — performance unexplained by finances alone.

Leicester City 2015–16: The Greatest Statistical Upset



Leicester City 2015–16	
Financial Rank	11th of 20 (PL)
Predicted pos.	11.5th
Actual pos.	**1st (Champion)**
Residual	−**10.5** positions
SD units	−**3.2** SD
Cook's Distance	**Highest in dataset**

4 of 50 champions (8 %) came from FR > 3. Napoli (FR 5), Lille (FR 4), Liverpool (FR 4) were within 1 SD of expectation. Only Leicester (FR = 11) is a genuine statistical outlier.

Robustness Checks: The Core Finding Survives Every Test

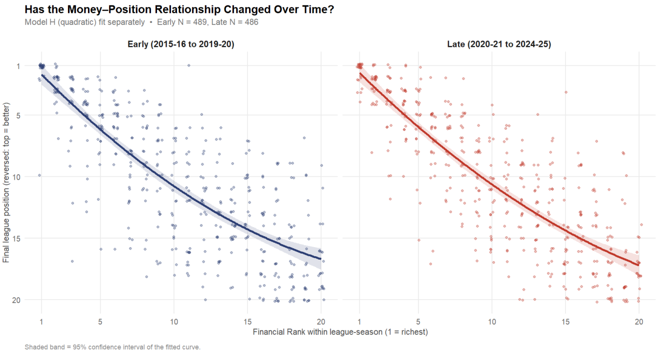
Test	Statistic	p-value	Verdict
Quadratic vs. linear (FR^2)	$F = 31.71$, 1 df	< 0.001	**FR^2 confirmed**
FR $\times$ League interaction	$F = 1.17$, 4 df	0.32	Effect uniform
FR $\times$ Season time trend	$F = 0.44$, 1 df	0.51	Stable 2015–25
Cubic term (FR^3)	$F = 0.20$, 1 df	0.65	Model H sufficient
H4 vs. H (League + Season)	$F(5) = 0.247$	0.941	Controls add nothing
Heteroskedasticity (BP test)	$\chi^2 = 45.5$	< 0.001	Present; HC1 SEs used

Models H2, H3, H4 all score higher AIC than Model H. Neither league fixed effects nor a time trend add explanatory power. Model H ($FR + FR^2$) is the correct and sufficient specification.

Statistical: Under HC1 robust SEs, the FR coefficient remains $p < 0.001$.

Plain English: The finding survives every specification test applied.

Extra Robustness: Splitting the Decade in Two

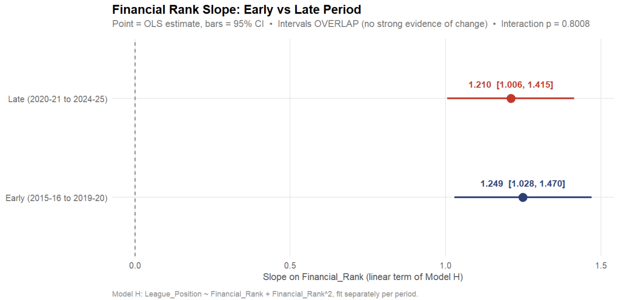


Model H re-fit completely independently on two five-season halves — if financial inequality were increasing (or shrinking), the two curves should visibly differ.

Period	R^2	FR slope
Early (15/16–19/20)	0.646	1.249
Late (20/21–24/25)	0.698	1.210

The two quadratic curves are visually almost indistinguish-able — confirming Slide 13's time-stability finding with a completely independent test design.

The Financial Rank Slope Barely Moved Across the Decade



Interaction $p = 0.80$

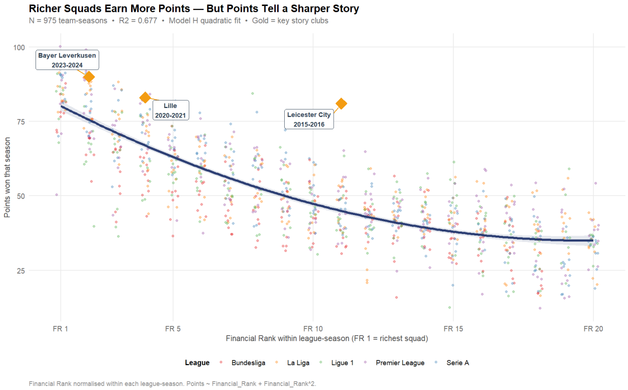
Financial_Rank × Period

— not remotely significant

- Early slope: 1.249, 95% CI [1.028, 1.470]
- Late slope: 1.210, 95% CI [1.006, 1.415]
- Intervals overlap almost completely.

Two independent designs — continuous and binary — agree: the relationship has not measurably changed since 2015–16.

Robust to Outcome Measure: Points Instead of Position



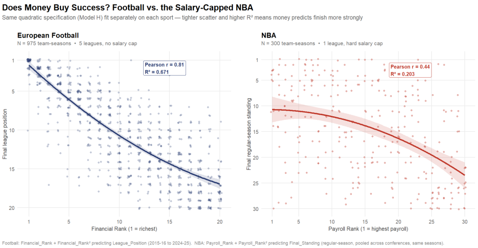
$$R^2 = 0.677$$

Points $\sim$ FR + FR^2 (vs. 0.671 using position; sign flips — FR is now **negative**, since more points is better)

- **Leicester 2015–16 still has the single largest residual** in the whole dataset.
- New story: rich clubs that *collapsed* — Chelsea, AS Monaco, Bayer Leverkusen.

Position or points — Financial Rank predicts either, and Leicester 2015–16 is the largest anomaly either way.

External Validity: Does Money Buy Success Outside Football?



Sport	N	r	R^2	Top-3
Football	975	0.81	0.671	92 %
NBA (cap)	300	0.44	0.203	50 %

A hard cap visibly weakens the pattern. Oklahoma City's 2024–25 title, won with the league's 23rd-highest payroll, would be a Leicester-level outlier in football; in the NBA it is routine.

Conclusions: What Ten Years of Data Tell Us

1. Money strongly predicts league position.
$r = 0.813$, $R^2 = 0.671$ with FR alone across 975 team-seasons ($p < 0.001$): **67 %** of positional variation explained.

2. The relationship is non-linear (diminishing returns).
FR^2 term: $F = 31.71$, $p < 0.001$. Marginal penalty: 1.19 positions per rank at the top vs. 0.63 positions at the bottom.

3. The effect is stable across leagues and over time.
FR $\times$ League: $F = 1.17$, $p = 0.32$ (uniform). FR $\times$ Season: $F = 0.44$, $p = 0.51$ (stable 2015–25).

4. Only one true exception in ten years.
4 of 50 champions from $\mathrm{FR} > 3$. Only Leicester 2015–16 ($\mathrm{FR} = 11$, residual $= -10.5$, $= -3.2\,\mathrm{SD}$) is a genuine statistical outlier.

Limitations

- Market value as quality proxy (injuries unmodelled)
- Managerial quality not captured
- Panel structure ignored
- Top-five leagues only

Financial dominance is universal, stable, and statistically powerful across all five leagues and all ten seasons. Leicester City 2015–16 is the exception that proves the rule.

Conclusions (continued): Extended Robustness Holds Up Too

5. Survives Every Extended Check

An independent Early/Late split (interaction $p = 0.80$) and a points-based re-fit ($R^2 = 0.677$ vs. 0.671) both confirm the core finding — and both still flag **Leicester City 2015–16** as the single largest statistical anomaly.

6. Generalises Beyond Football — Weaker Under a Cap

The NBA shows the identical quadratic pattern, but far more weakly: $R^2 = 0.203$ vs. 0.671; only 50 % vs. 92 % of titles to top-3 spenders. Evidence a hard salary cap loosens the spending-winning link.

Financial dominance is universal, stable, and statistically powerful — across all five leagues, all ten seasons, an alternative outcome measure, and even a different sport. Leicester City 2015–16 is the exception that proves the rule; a hard salary cap is one of the few things shown here to meaningfully dent it.

Thank You — Questions?

Money and Success in Football

Does Financial Strength Predict League Position?

$$r = 0.813 \quad R^2 = 0.671 \quad N = 975$$

Five leagues · Ten seasons · One answer: **yes, and very strongly so**

Data: ESPN FC + Transfermarkt
Period: 2015–16 to 2024–25
Models: OLS (A–H, H2–H4), Logit (E, J)
Contact: `mdmuktadir0611@gmail.com`

Key Numbers to Remember

- **76 %** of titles: won by FR = 1
- **92 %** of titles: from financial top 3
- **67 %** of position variance explained by FR alone
- **1** genuine statistical outlier in 10 years: Leicester City 2015–16
- **NBA check:** $R^2 = 0.20$ under a salary cap vs. **0.67** in football

Leicester City 2015–16: FR = 11, predicted 11th, won the league. Residual = -10.5 pos. = -3.2 SD. The model said impossible. They proved it — once in a generation.